00 / Post-event can consolidation

Newton 1.5.0 • Environment engineering report • Digital-only research and validation

The deposit_cans simulation: documented component qualification (solver v6, scene 7)

Two standard robot arms, a movable bussing tub and a freestanding bag-supported recycling station. The task is physical consolidation, not a scripted stream of cans into an invisible box.



Actual full 72-can scene at tick 7200 (30 s), solver revision bounded-shell-v6, scene revision 7. Recorded teacher data rendered without advancing or resetting the simulator. This endpoint alone does not establish persistent rest.

This dossier records the implemented environment, measured component checks, failed controls and demonstrated limits. Another policy solves the task. Assisted fixtures qualify physical interactions without claiming an autonomous solution or accuracy outside the documented range.

01 / The real workflow

Grounding: the complete action-group review and event-collection practice [S1–S6].

After an event, opened and drained aluminium beverage cans remain on a return counter and in a bussing tub. The robot must deposit the designated cans into the collection bag, preserve reusable cups and existing contents, avoid floor losses, keep the station intact, and return the empty tub. It chooses individual transfers, small batches or controlled tipping.

The action-group review covers 86 action occurrences, 86 distinct names, 12 leaf groups, 17 robot capabilities and 18 physics capabilities. The exact can action specifies a collection bag. Other leaves motivate supported release, reorientation, clearance and pouring. These taxonomy counts are not empirical task frequencies or proof that every hazardous-waste label is represented.

Inventory	Reference	High inventory
Targets in tub	16	16
Targets on counter	8	8
Pre-existing cans	48	120
Total physical cans	72	144
Standard arms	2 FR3	2 FR3

The setting is a venue service alcove: return counter, recycling frame, adjacent general-waste destination, reusable-item tray, tubs, wall protection and service-door surroundings. Bagged event collection is not asserted to be acceptable downstream municipal recycling; county guidance includes removing the collection bag at transfer.

Success is an outcome, not a prescribed sequence. No can crushing, bag installation, knot tying, beverage-fluid simulation or downstream sorting machinery is required. These exclusions prevent the benchmark from silently expanding into unrelated tasks.

02 / Objects at real scale

Published dimensions are separated from reconstruction and material estimates.

Object	Reference / implementation basis
Station [S2,S3]	1.016 m high; 635×457.2 mm footprint; 406.4×114.3 mm slot. System 3.62874 kg, lid 0.79379 kg. Fillets and wall details reconstructed.
Film bag [S4]	40×45in lay-flat reference; 38.1 µm film. Not a 40in inflated diameter. Nominal two-wall area 2.322576 m².
Bussing tub [S6]	389×506×125 mm envelope. 0.70 kg is a case-mass-derived estimate, not measured net mass. Recessed interior has physical walls.
Cans [S7,S8]	Standard 66×115 mm; slim 57×146 mm; tall 66×167 mm. Open ends, thin walls and thicker end regions.
Reusable cups [S21]	Hollow 75×132 mm tumbler reference. 77 g and wall/taper reconstruction are estimates.

Scene revision 7 includes an actual triangulated membrane with an open mouth, closed bottom, foldover cuff and assembled retention regions; a continuous gathered bottom (2.30743 m², 80.88 g); and a molded tub rim, recessed grips and bottom ribs. These are reconstructed physical geometries, not an exact eight-ply manufactured seam. The isolated material fixtures in chapters 4 to 6 use the earlier v4 solver revision and say so.

The station is freestanding with floor contact. The locked lid/frame assembly is a real initial assembly, distinct from a task grasp. The bag is not pinned to the world and does not surround a hidden rigid containment box. Each preloaded can contributes its own mass, contact and motion.

03 / Can mechanics

Fixed-topology thin shells; same surface for contact, deformation and rendering.

Mass is lumped from each triangle’s area, local wall thickness and aluminium density 2700 kg/m³. Body walls use 112–118 µm by type; end-region thickness 220 µm is an explicit reconstruction. This avoids both solid-aluminium can mass and density changes introduced merely to stabilise a solve.

The membrane uses the installed stable Neo-Hookean surface law. E=69 GPa, ν=0.33 and first-yield stress 220 MPa are researched alloy estimates. With surface shear µt and area coefficient λt, the energy can be expressed as $A[\mu t(I-2\ J)/2 + (\lambda t+\mu t)(J-1)^2/2]$, where $I=\mathrm{tr}(F^TF)$, $J=\sqrt{\det(F^TF)}$. Objective metric-rate damping is retained.

Bending energy is$\frac{1}{2}\ k_e\ L_e\ \mathrm{wrap}(\theta-\theta_0)^2$. The edge coefficient follows discrete-shell diamond geometry: $k_e=3L_e/(A_1/D_1+A_2/D_2)$, $D=Et^3/[12(1-\nu^2)]$. End and wall compliances are therefore not replaced by a single whole-can mean thickness. A dissipative rest-curvature update limits the elastic bending moment, allowing modest permanent dents.

The update is not full metal forming: no adaptive remeshing, tearing or fracture. Retained normal/over-grip comparisons and opposed-load release experiments below bound its demonstrated range; they do not establish arbitrary crushing accuracy. Published axial crush results were not copied into a radial finger-force threshold.

Digital constitutive check	Recorded result
Membrane force / energy derivative	Relative error 2.01e-08
Hinge force / energy derivative	Relative error 2.8e-08
Rigid translation control	No force change in the tested precise-coordinate fixture
Faster algebra against retained reference	512 hinges, damping included; force and curvature match after output rounding

04 / Measured dent and recovery

Material fixtures on the earlier v4 solver revision; explicit assistance and sensitivity, not full-scene qualification of the v6 solver.

These seven-second component recordings apply two opposed distributed loads to fixed material patches on the can wall. The force is ramped up, held, ramped down and removed; three final seconds are unforced. This is an explicitly assisted material fixture, not a robot grasp, real indenter or hidden task assistance.

Fixture	Peak / residual displacement	Final plastic work
Unloaded control	0.0003 / 0.0002 mm	0.000000 J
30 N per opposed patch	2.7743 / 1.9686 mm	0.034519 J
Same load; half timestep	2.7745 / 1.9687 mm	0.033408 J
Same load; 48 vs 24 segments	2.6879 / 1.0635 mm	0.054487 J

Nonrigid displacement is measured after removing the best-fit rigid motion. The zero-load control stays near numerical precision. The loaded shell retains a measurable dent after the external force is removed, with nonzero dissipated plastic work. An 80 N robot command alone was not used as proof of plastic deformation.

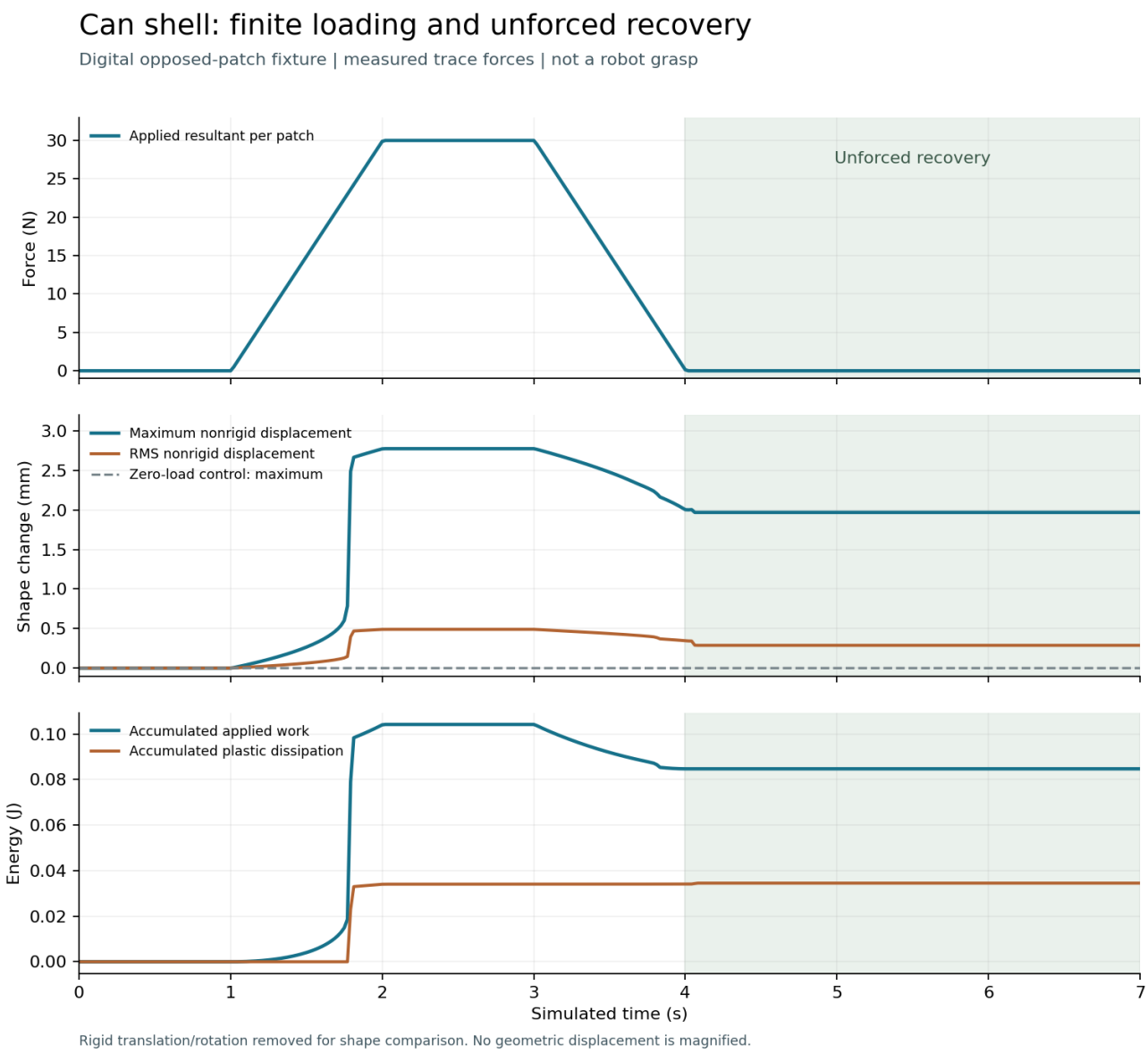
Halving the timestep changes final maximum dent by 0.0082%. Doubling circumference resolution changes it by 46.0%. This is not a clean constitutive mesh-convergence comparison: the vertex-selected fixture also changes its load support with the mesh. A patch-support check finds about 8.60 versus 12.88 mm nodal width. These outcomes demonstrate denting, but do not isolate material discretization error or establish mesh-independent dent depth.

The net fixture force is below 6.7 micro-newtons and net torque below 1.8e-8 N m in these runs, consistent with the intended opposed loading. These small net values do not mean the individual patch loads are small. Contact support, prescribed external work, elastic energy and plastic dissipation remain distinct quantities.

The v4 revision shares its bending, shell constitutive, geometry and precise elastic-force modules with the shipped v6 solver. Its contact, solver and integration code differ, so these recordings are constitutive evidence, not a substitute for coupled-physics qualification of the v6 solver.

05 / Actual loading history

v4 material fixture; plotted from recorded external forces and shell states.



Every curve uses retained samples from the opposed-load experiment and its zero-load control. The shaded interval has no applied patch force. No displacement is exaggerated in the geometry.

The can first deforms elastically, then buckles locally during the increasing load. It partially recovers as the load is removed and retains a residual dent. Plastic dissipation increases during irreversible deformation and stays nonnegative. Accumulated external work can decrease during unloading as elastic energy is returned to the fixture; it is not itself dissipated energy.

This shows a loading/unloading mechanism, not a calibrated prediction for arbitrary can crushing. The mesh/load-support limitation on the preceding page and the separate full-scene qualification both apply.

06 / Gravity, contact and impact

A 100 mm drop and timestep comparison on the v4 solver revision; real contact, no runtime support.

One can begins 100 mm above the actual counter, with zero initial velocity. Initialization is the only assistance. Gravity, deformable contact and stationary robot commands determine its subsequent motion. The component omits the other cans, so it does not establish full-pile stability.

Quantity	dt = 1/240 s	dt = 1/480 s
First recorded counter contact	0.141667 s	0.141667 s
Point-body gravity reference	0.142809 s	0.142809 s
Peak recorded nonrigid displacement	0.02238 mm	0.01637 mm
Final kinetic energy	1.56e-11 J	1.09e-11 J
Final plastic dissipation	0 J	0 J

Both recordings reach physical contact at about 0.142 s and finish five simulated seconds without a fault. The reference excludes shell deformation and contact thickness; it is a comparison, not an exact target imposed on the simulation. This modest drop is predominantly elastic in the model, not falsely labelled as a permanent dent.

The recorded deformation peak differs between timesteps. Geometry snapshots are sampled more sparsely than physics, so these peaks are sampled observations rather than guaranteed instantaneous maxima. Contact time, final energy and geometric response are reported separately instead of claiming perfect timestep invariance.

Energy records separate membrane, bending, kinetic, gravity and plastic terms. They omit some contact/rigid dissipation and are not presented as a closed-system energy-conservation proof.

07 / Bag load paths

A thin film can bend easily while resisting in-plane stretch.

The film model uses 38.1 µm thickness, 920 kg/m³ density, E=200 MPa andν=0.45. The areal density is 0.035052 kg/m², in-plane modulus Et=7620 N/m and bending rigidity about 1.16×10^{-6}N·m. These are comparator-based polymer estimates, not a measured constitutive curve for the selected recycled blend [S12].

Assembled retention springs connect actual film patches to coordinates moving with the finite-mass frame. The v6 path uses implicit retention with an opposite frame reaction. Retention forces and attachment errors are recorded each substep. Massless assembly endpoints are excluded from material self-contact; real bag/can contacts remain enabled.



Actual tick 7200 bag-bottom view. Material-following print, film shape and individual cans are rendered from the physical mesh. The endpoint image is not substituted for the time-series rest check.

No puncture or rupture model is claimed. Ordinary loading must nevertheless remain stable and non-penetrating. A numerical wall crossing is a fault, not a supported way to put a can inside. Floor support is permitted only through actual bag/floor/can contact, never an invisible shelf.

08 / Robot contact and assistance

Commercial arm/gripper specifications; finite effort; assistance disclosed.



Zero-step initial fixture: both open fingers straddle the can. The robot is initialized here before states are finalized; this is not a runtime teleport. The picture does not establish a successful grasp.

The FR3 kinematic chain comes from the recorded manufacturer-style URDF asset. The WSG50-110 mounting, padded fingers and compound inertias are authored engineering reconstructions, not a factory-certified FR3 package. The nominal gripper envelope and operating range follow the product table/manual [S9, S10].

Control	Implemented meaning
Arm motion	Bounded joint targets; manufacturer effort limits enforced by the integrated bounded native VBD motor.
Gripper	110 mm total opening; explicit MOVE positioning or GRIP at 5–80 N total. GRIP closes at 50 mm/s relative, 25 mm/s per jaw, with bounded effort.
Assisted fixture	Privileged can-centre IK and finite servo targets. No can forces, attachments or runtime pose writes.
Path claim	IK convergence alone is not collision-free motion or a reached physical pose.

09 / Numerical safeguards

Why the solver is built this way. A finite state alone is not a physical pass.

Safeguard	Reason / remaining boundary
Global contact-coupled correction	Native local iterations alone do not converge the stiff can, and impact energy grows. The correction addresses stiff coupled forces; it is not IPC dynamics or a blanket convergence guarantee.
Wrapped hinge angles	Principal wrapped elastic/plastic angle differences avoid the signed-angle discontinuity; this alone does not prevent impact failure.
Symmetric self-contact filtering	A one-sided native topology exclusion would omit reverse edge queries. Symmetric topological filtering keeps the tested force balance without disabling disconnected contacts.
Double shadow coordinates	Independently rounded world translations would inject metal strain. Double shadow coordinates preserve sub-ULP motion; contacts and images use the same rounded public coordinates.
Sequential rigid-body updates	Joint coloring alone does not cover dynamic prop contacts. Conservative sequential body updates avoid same-color divergence.
Separate checkpoint rest storage	Rest storage is kept apart from the contact detector’s mutable vertex buffer. A fresh model/store restores all captured numeric bytes.

The bounded-shell-v6 solver uses 80 native iterations, 40 rigid iterations, 64 inner Krylov iterations and a global correction every 8 iterations at dt=1/240 s. CUDA graph capture changes dispatch, not the number of physics steps. An optional same-rule work-evaluation shortcut is explicit in each run configuration. No hidden speed cap, added object force or changed material modulus is used to meet the cost target.

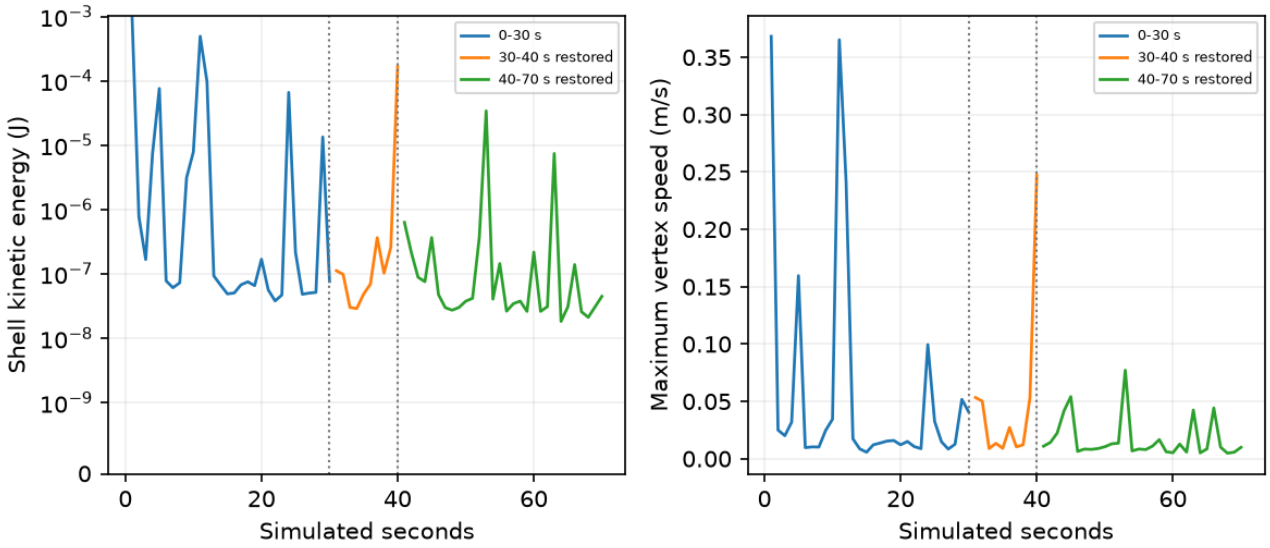
The global correction changes numerical solution and arithmetic, not the researched mass, modulus or contact force. It uses a positive approximate search matrix, global contact-coupled Krylov reductions, rigid-mode preconditioning and a bounded line search. Some linear residuals still miss tolerance; no blanket convergence claim is made.

Adaptive native collision re-querying keeps the collision search bound from capping speed: without it, a contact-free can starting at 4.8 m/s is clipped to 2.04 m/s in a four-tick free-flight check. A 160-tick floor-impact fixture remains finite, with minimum shell height 0.478 mm and no energy growth above its initial 0.14828 J. This is a targeted component check, not full-scene acceptance.

The material force is integrated along candidate steps with three-/five-point work estimates. Requiring positive estimated work beyond quadrature disagreement is a numerical acceptance rule, not a theorem of energy conservation for friction or a replacement for physical rollout checks.

10 / Rest, rolling and pile rearrangement

Unforced baseline through 70 s; explicit checkpoint boundaries at 30 and 40 s.



Actual recorded summaries; logarithmic energy scale near rest. The dotted lines disclose checkpoint restorations. These three segments are not presented as uninterrupted video.

Recording	Captured substeps / faults
0-30 s	7200 captured; 0 recording errors; 0 evictions; run error=None.
30-40 s	2400 captured; 0 recording errors; 0 evictions; run error=None.
40-70 s	7200 captured; 0 recording errors; 0 evictions; run error=None.

Eight independent counter cans show zero stored vertex-coordinate change over 60-70 s. They remain ordinary positive-mass contact objects, not frozen props. The robot transfer fixture demonstrates that a counter can can subsequently be grasped and moved.

The loaded pile does not pass the stricter whole-pile rest audit (physical_rest_pass=False). In its final 10 s, maximum shell kinetic energy is 7.403e-06 J. The final-second maximum individual vertex speed is 166.72 mm/s; this is not centre-of-mass speed or sustained drift. Bag vertices move up to 2.927 mm over the last 10 s; rigid origins move up to 0.179 mm.

There are no floor losses, ejected preloads, invalid geometry or surface-path faults in these segments. An instantaneous settled flag is not used as proof that every can has stopped. No extra stabilization force, frozen can or successive rest extension was introduced to turn the audit into a pass.

11 / What the late motion actually is

Read-only mass-weighted analysis of the recorded meshes and contact loads; zero new physics steps.

65-70 s	COM displacement (x, y, z) mm	Rotation	Shape RMS
can_001	+0.158, -9.877, +0.051	19.638 deg	0.299 um
can_003	+0.199, -9.542, -0.055	14.453 deg	0.547 um
can_015	-1.004, -1.418, -9.029	8.433 deg	0.445 um
can_067	+0.214, -0.119, -0.863	0.487 deg	0.060 um
can_068	+0.807, -0.176, -0.971	0.450 deg	0.041 um

can001 moves about 9.88 mm sideways while rotating 19.64 degrees. Its nominal 28.5 mm radius gives a rolling arc of about 9.77 mm, and the fitted shape residual is 0.299 micrometres. This is consistent with rolling of a nearly undeformed can, not an expanding shell or arbitrary pose translation. Recorded contact with can003 disappears by 68 s, after which the remaining tub normal force is approximately the can weight.

The motion is intermittent and changes bodies: can008 moves about 17.18 mm sideways during 60-65 s and less than 0.1 mm in the following window. can015 descends 9.03 mm in 65-70 s. Individual one-second COM samples include a 22.58 mm/s packing transient at 63 s, while most sampled maxima are below 2 mm/s. These sparse velocity samples do not bound all sub-second peaks.

Total can gravitational potential decreases from 3.618871 to 3.582307 J over 40-70 s (drop 0.036565 J), and at each five-second boundary. The evidence supports gravity-driven rearrangement and rolling rather than global spontaneous energy growth. It is not a closed energy balance: film, contact and internal-energy terms are not included in this sum.

Contact loads are recorded final-iterate numerical forces, not measured sensors. Directed edge records expose only the acted-on edge vertices; the report does not invent an opposing-body identity when that record does not provide it. The exact duration of real pile settling is not established. The supported claim is stable, finite, contact-driven packing with stationary independent objects, not a perfectly motionless loaded pile.

The complete 40-70 s segment has 361 frames at 12 fps. Restoration at tick 9600 is captioned; no interval is hidden.

12 / High-inventory loading

144 physical cans: 24 targets plus 120 existing cans. Thirty seconds of unchanged robot targets.



Actual tick 7200 bag-bottom close-up. The initial stacked preload has packed downward into a disordered pile. The film, printed material coordinates and all individual cans come from the recorded physical state.

High-inventory record	Measured result
Physical duration / history	7200 ticks / 30.0 s; 7200 retained samples; no capture errors or evictions
Physical fault / losses	None; valid geometry, retained station, no floor-can or preload loss
Late bag / station motion	5.536 mm maximum bag-vertex motion; 0.117 mm maximum station-origin motion over 20-30 s
Final shell kinetic energy	6.74931e-08 J
Strict whole-pile stationary audit	False; continuing packing is not hidden

The largest final-window can motions are predominantly downward: can079 descends 27.87 mm and can103 descends 21.69 mm during 25-30 s. Their best-fit nonrigid RMS is 0.694 and 0.297 micrometres respectively. These are translations/rotations of nearly unchanged cans, not large fabricated shell deformation. Independent counter cans remain unchanged at stored precision.

Can gravitational potential drops from 9.570962 J at initialization to 6.583278 J at 30 s; it also decreases through each five-second boundary. The final five-second decrease is 0.017821 J. This supports gravity-driven packing. The sum excludes film and contact energy, so it is not a closed-system conservation proof.

The audit still records transient late motion: peak shell kinetic energy 0.000609 J over 20-30 s and final-second maximum can-vertex speed 73.11 mm/s. These peaks are not equivalent to the much smaller final kinetic energy or a constant COM drift. No extra damping, frozen preload or quiet endpoint substitution was added.

Both mounted 1280×720 wrist images were captured every 0.5 s throughout the 30 s run. The run saved an endpoint checkpoint and released both simulator and renderer. Its 739.64 wall/sim ratio includes concurrent qualification and rendering; it is not an isolated latency measurement.

13 / Physical grasp, transfer and release

Full 72-can inventory; one fixed assisted component sequence, not a task-solving policy.



Actual tick 1308 close-up: the finite-effort robot holds can016 above the counter. Both contact pads and the displaced can are visible. This is recorded geometry, not a pose-welded illustration.

Component stage	Recorded physical result
Initial assistance	Open robot initialized around can016 before the first state; no object repositioning
Grasp / lift	10 N total GRIP; can bottom rises 142.998 mm; first endpoint error 1.791 mm
Transport	50 mm/s commanded path; final TCP error 2.972 mm; no corrective retries
Release / observation	1 s opening, followed by exactly 30 s of unchanged targets
Completed physical time	15564 substeps / 64.85 s
Recording	15564 retained samples; zero capture errors or evictions

The counter can is lifted, carried over the mouth, physically released and retained inside the bag. The remaining 23 targets are deliberately not solved. No external object forces, attachments, runtime pose writes, resets or success-seeking retries are used. Privileged can-centre IK and the prescribed finite robot commands are disclosed assistance.

All recorded physical geometry, station, reusable-item and loss checks remain valid. The strict whole-pile stationary audit is false: can vertices move up to 13.366 mm over the last 10 s; peak late shell kinetic energy is 0.00013473 J and final-second maximum can-vertex speed is 96.88 mm/s. Those are transient maxima, not the final energy or sustained target-centre speed.

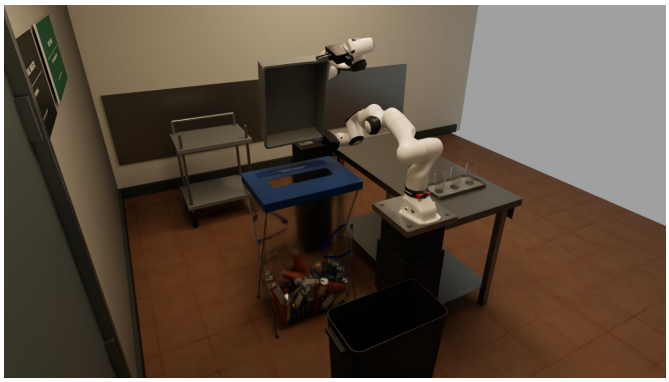
In the final 4.996 s, the deposited can centre moves 4.566 mm, including -3.536 mm vertically. Its fitted centre trend is 0.899 mm/s, down from 1.123 mm/s in the preceding window. This is continuing slow packing, not a motionless-pile claim; no further rest extension or stabilization was added.

14 / Bimanual tub contact and discharge

A completed finite-effort component recording, followed by 30 seconds of observation.



Tick 240: physical fingers support the tub. Three component cans and 48 preloads are present. Initial placement assistance is disclosed.



Tick 13008: tipped tub empty; three cans inside the bag. Exact retained states; zero additional physics steps.

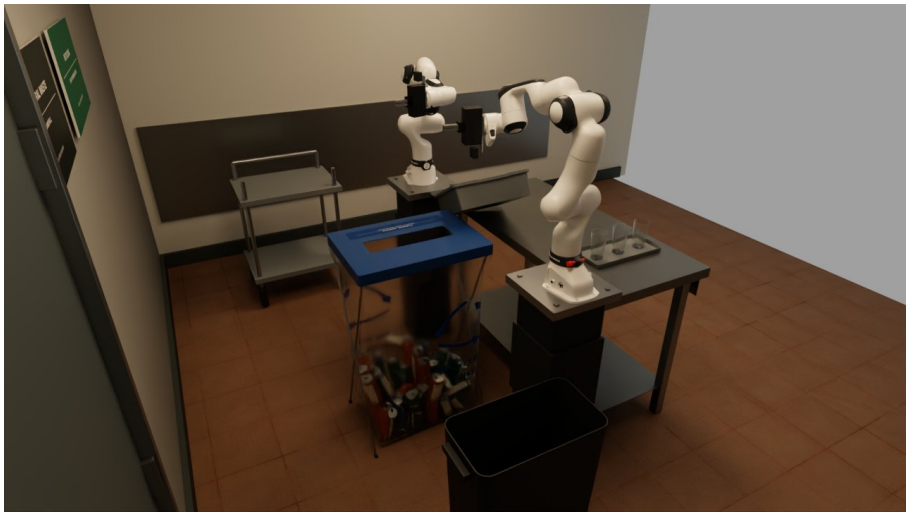
Component record	Measured outcome
Physical duration	13008 substeps / 54.20 s
Actuation	40 N total per gripper; finite joint targets; approximately 102-degree tip
Targets inside / floor losses	3/3 inside; 0 floor losses
Post-tip observation	30 s; no applied can forces or resets
Trace	13008 samples; 0 capture errors; 0 evictions

No can/tub welds, runtime pose writes, scripted can emission or external object forces are used. The arms continue holding the tub during post-tip observation. The tub is not returned to the counter, so this is deliberately not a whole-task solution. A short wall-clock pause for an isolated cost measurement preserved the same in-memory trajectory; no simulation time was skipped.

At the final endpoint all three component cans are released, the station is upright, retention is intact, and no preloaded can is ejected. The final approximately five-second window has at most 1.622 mm net can-centre movement, with the largest fitted centre trend 0.319 mm/s. This is post-transfer settling evidence for this recording, not a claim that every initial full-inventory stack is motionless.

15 / When the grip is insufficient

Negative component control: a failed grasp is not a corrupted simulator.



Actual final state at tick 600 (2.5 s). The tub has slipped away and caught on nearby equipment; the arms remain near their unchanged targets. The trace, not this camera angle, establishes the two floor losses.

This starts from the same recorded initial tub/can and robot placement as the stronger-grip fixture. Only finite robot commands follow. Each gripper is commanded to close with 5 N total effort; no object forces, welds, runtime pose writes or stabilizing attachments are applied.

Recorded quantity	Result
Duration	600 substeps / 2.50 s
Fixture outcome	Unsupported hold fails; tipping is not executed
Tub motion	650.158 mm translation; 0.530018 rad rotation
Left / right TCP target error	0.092 / 1.016 mm
Simulator / recording	Geometry valid; 600 recorded samples; no capture errors or evictions

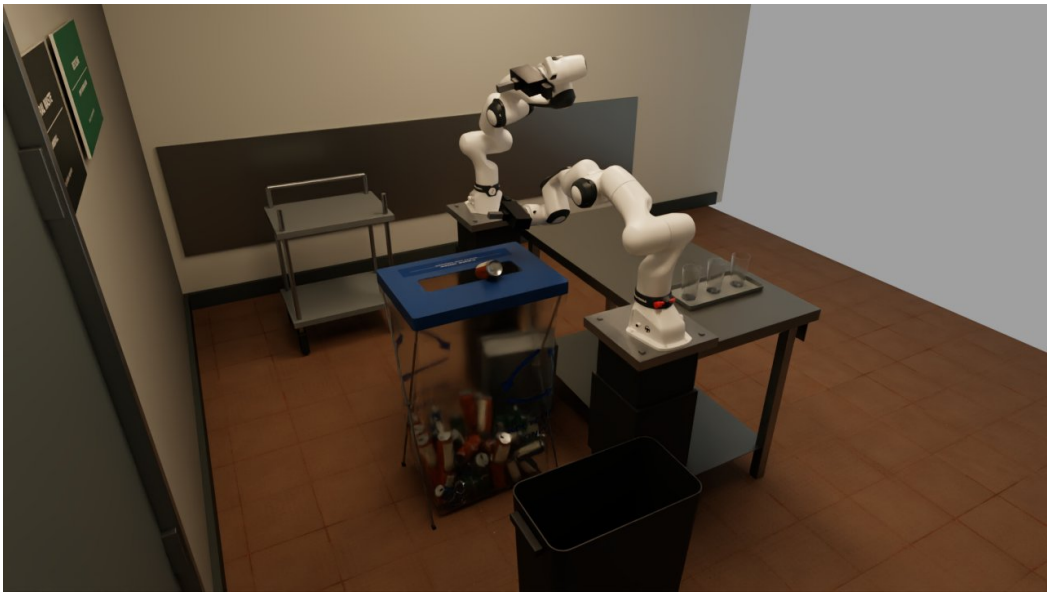
The fingers no longer contact the tub at the endpoint. This is useful evidence that inadequate grasping can fail while robot targets remain stationary. It does not establish a calibrated slip-force threshold or make the exact transient friction mechanism a measured fact.

The 5 N and 40 N records come from different source revisions with the same physical parameters and plan; optional scheduling and batched-force features are disabled in both. They are comparable by source review, not by identical archive hashes or deterministic GPU trajectories.

The complete video has 121 frames at 48 fps (2.521 s). Initial assistance and the failed fixture are captioned. Its frames and this final frame show real separation of the tub and fingers; no failure interval is removed.

16 / Bad aim and a faster command

One fixed attempt of each negative control. Failed outcomes are preserved, not optimized away.



Faster-tip endpoint: the tub has slipped out of the grippers; one can rests on the inlet rim. A second remains supported by the fallen tub; only one is contained. An endpoint image alone does not certify the full motion.

At exactly 3 s after tip commands	Inside	Geometry / history
Slower 102-degree tip	3/3	valid=True; 13008 substeps, no capture errors
Same endpoint, twice target rate	1/3	valid=True; 3924 substeps, no capture errors
80-degree / offset-outlet control	0/3	valid=True; 4680 substeps, no capture errors

The faster control follows the same original waypoint plan with every second waypoint selected: 10.85 s of target motion versus 21.7 s. Gripper effort and physical parameters are unchanged. At the matched three-second observation, the slower sequence contains 3/3 and the faster 1/3. This is an observed pair of outcomes, not proof that doubling speed deterministically causes failure in every GPU replay.

The actual fast-control contact trace places can002 on the rounded-slot lid with a summed upward last-iterate force of 0.168209 N. It is neither below the mouth nor inside; the verifier refuses to count it. can001 remains supported by the tub outside the bag, while can000 is contained. This checks a meaningful rim-perching failure, not merely a finite-state guard.

The off-target control stops at 80 degrees with an outlet offset of −0.18 m. It completes 19.5 s and leaves 0/3 inside, without floor-can loss, ejected preloads or a numerical fault. No corrective waypoint, extra attempt or success-seeking reset follows. Its result demonstrates that the environment permits unsuccessful finite robot commands.

Both negative controls use three component targets plus 48 pre-existing cans, not the full 24-target task. Their post-tip observation is 3 s; the slower record additionally retains its full 30 s observation. Comparisons use the common 3 s interval, not unequal final settling times. The records come from different source revisions, and contact trajectories are sensitive.

17 / Verifier, history and recovery

Outcome geometry and continuity are separate from process health.

Containment uses the closed-bottom bag surface plus a virtual mouth cap. That cap is only a verifier construction: never rendered, never a collider. Can vertices and face centres must be inside and below the inner rim. The verifier also requires release, settling, an upright retained station, returned tub and preserved reusable items. Floor loss and ejected preloads are latched.

A winding test alone cannot rule out tunnelling. IPC Toolkit 1.6.0 checks the linear path between every completed substep for can/bag crossings and bag self-intersections. It does not replace Newton dynamics. The default 1 µm CCD tolerance can conservatively reject a path; the fault description distinguishes that from proof of a specific wall crossing. Can/can interactions remain physical Newton contacts, outside this containment-only query.

The v6 HTTP probe constructs the full 72-can environment and advances exactly one tick. Owner/restricted authentication, read-only observations, mounted PNGs, invalid-action non-mutation, idempotency, checkpoint save/list, pause rejection and zero-tick resume pass. The managed GAR launch and the dashboard in Chromium also pass: connect, exactly 20 requested ticks, pause rejection, refresh without advancement, resume, idempotency, restricted-key isolation, invalid-key rejection and disconnect. The temporary server is stopped afterward.

Check	Evidence
Reads / invalid requests	No state mutation in the 72-can HTTP probe.
Privileged action trace	1 sample, 0 capture errors, 0 evictions; completeness=True.
Longer telemetry	0–30 s: 7200/7200; 30–40 s: 2400/2400 substeps; no recording errors or evictions.
Durable control state	Real mode-change/restore check preserves native arrays, including gains, drive modes, effort and velocity targets.

Per-substep history contains object/robot poses and velocities, actuator targets, GRIP state, rigid and shell contacts, individual self-contact contributions, plastic history, retention load and residual diagnostics. Forces are numerical final-iterate evaluations, not measured force sensors or a trace of every internal solver iteration. Storage summaries expose sampling, missing ticks, failures and eviction. Restricted keys cannot retrieve privileged history.

18 / Observations and operating contract

Actual managed GAR launch and role-separated HTTP/dashboard checks.

Interface	Restricted policy	Privileged operator / teacher
Task	Natural-language objective	Target identities and detailed outcome
Cameras	Mounted wrist RGB	Mounted plus diagnostic views
State	Joint / gripper proprioception	Full object and material state
Geometry	Not exposed	Physical surfaces, assets and local frames
History	No hidden teacher trace	Per-substep state/contact trace with gaps
Controls	Finite joint, gripper and Cartesian commands	Same physical actuation plus explicit administrative operations

Commands advance exact tick counts. Reads, renders, geometry retrieval and variation selection advance no physics. Reset and restore are explicit continuity changes, never invisible recovery. Invalid actions must be rejected before mutation; numerical faults preserve completed tick/history and make the backend not-ready.

The two wrist cameras use ideal RGB pinhole projection on collision-bearing D405-envelope mounts. This is not a depth camera implementation or a calibrated sensor-noise model. An initially obstructed task view is not itself an environment defect; invalid mounting, missing geometry or stale image/state pairing would be.

19 / Cost and physical acceptance

The cost limit is 200 wall-clock seconds per simulated second.

Identical camera/history workload	72 cans	144 cans
Physics and control	485.11 s	749.45 s
Per-substep verifier / CCD	64.50 s	134.95 s
Teacher recording	31.50 s	59.32 s
Both mounted RGB cameras	20.29 s	28.38 s
Operating wall / simulated second	120.28	194.42
Total job elapsed	639.61 s	1070.89 s
Wall / sim including startup/finalization	127.92	214.18
Recorded workload	1200 ticks / 22 PNGs	1200 ticks / 22 PNGs

This run uses the CUDA-graph-captured v6 solver and the explicit unchanged-rule work shortcut. It includes full teacher history, actual geometry checks and both 1280×720/32-sample wrist images every 120 ticks. No requested workload or material is reduced. Startup and finalization make total elapsed 639.61 s. A five-second cost measurement does not establish 30-second physical rest.

The 144-can measurement uses the same configuration as its completed 30 s record. It is the only simulation job on a shared host; other services stay active and no time is subtracted for them.

Both operating ratios are within the 200x limit. The high-inventory five-second job costs 214.18x when startup and finalization are included, so it is not under 200x all-inclusive.

20 / Qualification scope and recordings

Component evidence, negative controls and explicit limits. No autonomous task policy is required.

Physical / package evidence	Recorded boundary
Single-can GRIP / transfer / release	Completed; 30 s post-observation; remaining 23 targets not solved
Tub contact / discharge / negative controls	Normal, weak grip, fast and off-target records retained
72/144-can unforced behavior	Finite contact-driven packing; strict whole-pile stationary checks remain false
CLI / HTTP / auth / dashboard	Actual no-override GAR launch and Chromium controls checked
Continuous single-transfer export	1558 frames; tick0–15564; exact tick timestamps; actual playback checked

The full single-transfer MP4 runs 64.854 s, with 1558 frames and maximum tick/timestamp discrepancy 3.33e-07 s. It renders the stored states from the beginning, with no physics restart or splice.

The package passes 53 CPU contract/lifecycle checks and the no-override GAR/Chromium probe: exactly 20 requested ticks execute, and owner/restricted isolation and pause/resume controls pass. Neither CPU tests nor dashboard checks replace the physical evidence above.

Digital-only constraints do not require laboratory access. Manufacturer specifications and defensible estimates are the research inputs. Known model limits are fixed-topology shells without tearing, approximate plastic denting and an uncalibrated film/retention reconstruction. Other seeds, arbitrary object crushing and every possible robot command are not certified by these fixed component checks.

21 / Sources and assets

Primary sources and asset origins.

The sources distinguish manufacturer claims, accessible paper sections, estimates and implementation deductions. The sources this report cites follow; no product advertising photograph is used as simulation evidence.

S1: Ramsey County event collection instructions

S2: ClearStream SingleStream Blue system specification

S3: ClearStream blue wide-mouth recycling lid, SKU 2WMROB005

S4: ClearStream selected clear/blue-print bag

S5: ClearStream bag installation instructions

S6: Cambro Poly Cambox specification

S7: Wang et al., empty-can experiments, Materials 17 (2024) 2655

S8: Shining Packaging 330 mL standard can specification

S9: Franka Research 3 datasheet R02212 v2.6

S10: Weiss Robotics WSG-series product data

S12: PTT film-grade comparator data

S21: Cambro Camwear 1200CW tumbler

Bending: Grinspun et al., Discrete Shells

S25: Stable numerics for finite-strain elasticity

S26: IPC Toolkit maintainer source

Asset provenance is recorded in the environment’s ASSETS.md. The installed FR3 URDF and meshes trace to franka_description revision d9841cf with the bundled Apache 2.0 license. The Poly Haven anti-skid floor maps are CC0, with original URLs and hashes recorded with the wiping assets. Gripper adapters, reconstructed scene geometry, fictional branding and film shaders are authored approximations, not manufacturer-supplied CAD or product photographs.